PLANRS ACADEMY

Science Art Academy: Grade 3 Chemistry • The Secret Life of Ink

Session 1: The Great Colour Race

A Mysterious Note!

Ananya found a secret note in the classroom written with a black sketch pen. She thought black was just one colour, but what if she's wrong? What if black ink is hiding a whole team of colours inside? Today, we use science to uncover the truth!

Your Science Art Goals

Today, we will become "Art-Scientists" to achieve these goals:

- **Understand Mixtures:** Learn that some things are a "team" of many parts.
- **Define Chromatography:** Discover the method used to separate hidden parts.
- **Perform the Experiment:** Watch the "Colour Race" happen on paper.
- **Become a Science Artist:** Use chromatography to create beautiful patterns.

THE QUICK GUESS:
What colours do you think are hidden inside a black sketch pen? Red? Blue? Yellow? Let's make a prediction before we start!

What Is a Mixture?

In science, a **Mixture** is when two or more things are put together but not permanently joined. Ink is a perfect example of a hidden mixture.

The Secret Team: Black ink looks like one colour, but it is actually many different colours hidden together like a team!

The Fruit Chaat Analogy

Think about a bowl of Mixed Fruit Chaat:

- It looks like one tasty snack.
- But inside, you can find separate pieces of Apple, Banana, and Papaya.
- Black ink is just like a "chaat" of different colours!

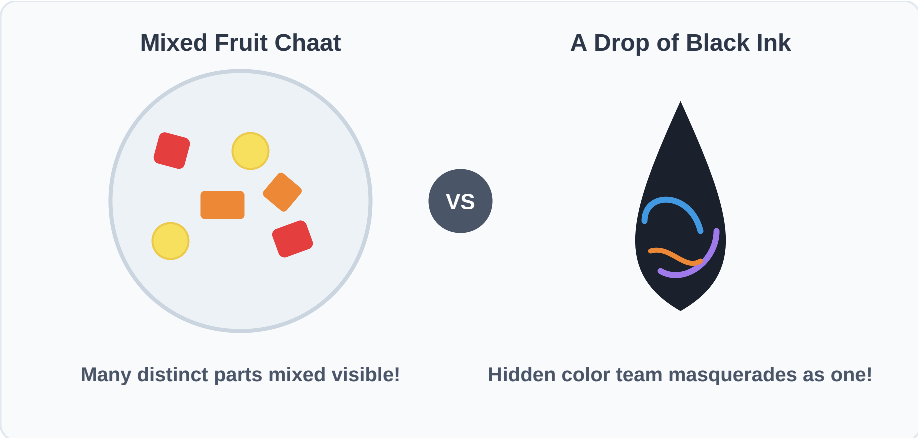

Figure 1.1: The Mixture Concept — Visible Components vs. Hidden Sub-parts

The Great Colour Race

If ink is a mixture, how do we separate the colours? We use a magic trick called **Chromatography**.

> **Definition:** Chromatography is a way to separate a mixture by making its parts move at different speeds through a material.

🌧️ **The Rainy Day Race**

Imagine a race on a rainy day in your colony:

💧 When the ground gets wet, some children run very fast.

💧 Some children run a bit slower on the slippery ground.

💧 In chromatography, water helps the colours "run" up the paper.

Fast colours reach the top first, while slow ones stay behind!

The Experiment: Uncovering the Secret

Let's set up our own Colour Race. You will be amazed at what you see!

What You Need:

- A black sketch pen (water-soluble).
- A strip of tissue paper or filter paper.
- A small drop of water.
- A flat surface (newspaper or table).

ART-SCIENTIST TIP:
Different brands of black pens use different colour "teams." Try two different black pens to see if the results are the same!

Step-by-Step Magic

1. **The Starting Line:** Draw a thick black line or a big dot near the bottom of your paper strip.
2. **Add the "Runner":** Place a small drop of water directly on or just below the black ink.
3. **Watch the Race:** Wait for a few minutes. Watch as the water starts to climb up the paper, carrying the ink with it!

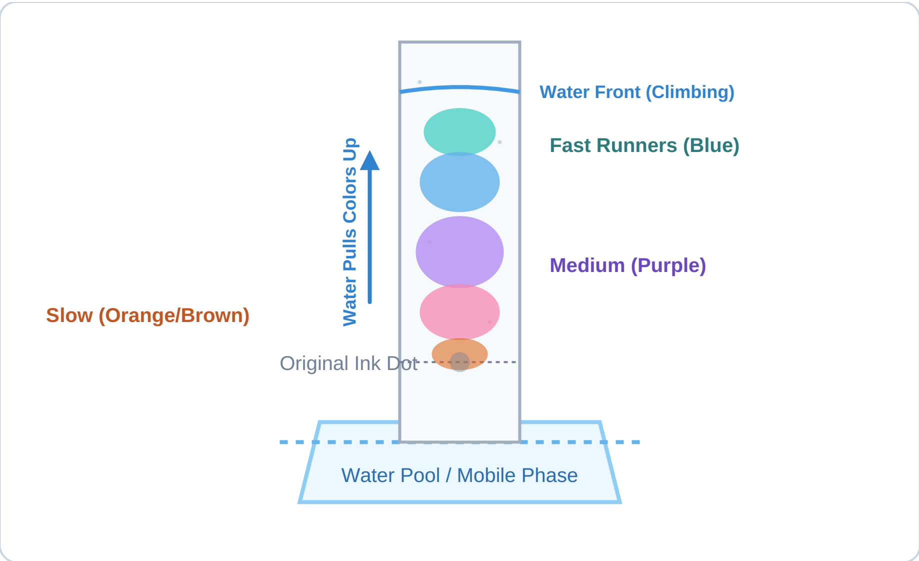

Figure 1.2: Chromatography Race — Component Separation Over Time

What Do You See?

As the water climbs, the "Black" ink will start to disappear. In its place, you might see a beautiful band of:

- **Bright Blue** & **Deep Purple** near the higher zones.
- **Warm Brown** & **Soft Orange** resting closer down.

The Science Secret:
The colours that you see at the top of the paper are the **"Fast Runners."** They love traveling with the water. The colours that stay near the bottom are the **"Slow Runners."** They like holding onto the paper tightly!

Interactive: Hidden Colours Challenge

Can you guess which colours are hidden in these other sketch pens?

1. **THE GREEN PEN:** Which two primary colours are hiding here?
   Your Guess: ____________________ and ____________________
2. **THE PURPLE PEN:** Which two colours make up the team for Purple?
   Your Guess: ____________________ and ____________________
3. **THE ORANGE PEN:** Which team is hiding here?
   Your Guess: ____________________ and ____________________

Be a Science Artist!

Chromatography is not just for scientists—it's for artists too! Use your separated colorful strips to make unique bookmarks or custom greeting cards.

Lesson Recap

- **Mixtures:** Black ink is actually a hidden "team" of many separate colours.
- **Chromatography:** We can separate these colours using a water climbing trick.
- **The Race:** Different colours move at different speeds on paper due to water pull.

STAY CURIOUS!

Next time you use a sketch pen, remember the hidden team of colours inside!